ANNA RUTH TAYLOR

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EDUCATION

Lunar and Planetary Laboratory, University of Arizona *Aug 2023 – Expected Spring 2028*
Ph.D. in Planetary Sciences · Minor in Astronomy · Advisor: Dr. Tommi Koskinen · GPA: 4.0

North Carolina State University *Aug 2019 – May 2023*
Honors B.S. in Physics · Minors in Mathematics and Computer Science · GPA: 3.97

PROFESSIONAL EXPERIENCE

Lunar and Planetary Laboratory — Research Assistant *Aug 2023 – Present*
University of Arizona · Advisor: Dr. Tommi Koskinen

- Investigating HeI triplet absorption at 1083 nm in hot escaping exoplanet atmospheres to understand transit depth trends and connections to mass loss.
- Running Fortran/C++ atmospheric modeling codes; analyzing output with Python.

Post Road Foundation — Researcher *Feb 2022 – Aug 2023*

- Ran and analyzed energy modeling simulations using NREL's ResStock housing stock code.

North Carolina State University — Astrophysics Researcher *Apr 2020 – Aug 2023*
Advisor: Dr. John Blondin

- Studied shocks and gas flow in astrophysical objects on a stellar scale using the Fortran code VH-1.
- Analyzed and visualized data from computational binary star models with Python and Ensight.

NASA/GSFC Internship — Astrophysics Researcher *Jun 2022 – Feb 2023*
Advisor: Dr. Sarah Peacock

- Computed photosphere+chromosphere models and UV-to-NIR synthetic spectra for FGKM-type stars using the PHOENIX atmosphere code.
- Analyzed and visualized synthetic spectra with Python and IDL; first-author publication accepted in ApJ.

Women in Physics Club — Vice President *Aug 2022 – May 2023*

- Organized career talks, social gatherings, and panel discussions; provided resources to undergraduate women in physics.

Senior Design — Critical Lead *Aug 2022 – Dec 2022*

— Created a precise positioning system for a lead gamma-ray collimator in a vertical plane; managed timeline and recorded design notes.

AWARDS & ACHIEVEMENTS

- **GPSC Travel Grant (2026)** Awarded by the University of Arizona Graduate and Professional Student Council to support international conference travel.
- **Zonta International Amelia Earhart Fellowship (2025)** Global fellowship for women in aerospace and space sciences; chosen as the Sharon Langenbeck Fellow.
- **Galileo Circle Scholarship (2024, 2025)** Awarded by the UA College of Science to exceptional graduate students demonstrating academic excellence and research achievement.
- **NSF GRFP Honorable Mention (2025)** Recognized by the National Science Foundation for research potential and broader impacts.
- **Leif Erland Andersson Graduate Student Award for Service (2025)** Recognized for contributions to mentoring, outreach, and service within the department.
- **Rodney I. McCormick Award (2023)** Awarded in recognition of research accomplishments as a physics undergraduate.
- **John Mather Nobel Scholar (2022)** Travel award recognizing outstanding undergraduate researchers.
- **NCSU Envisioning Research Contest Winner (2021)** Won the undergraduate video/interactive category for “Wind Driven Accretion onto a Black Hole.”

PUBLICATIONS

- **Taylor, A.R.**, Koskinen, T., Huang, C., Arfaux, A., & Lavvas, P. 2026, ApJ — “Helium escape in context: Comparative signatures of four close-in exoplanets.” doi:10.3847/1538-4357/ae41b5
- **Taylor, A.R.**, Koskinen, T., Argenti, L., Lewis, N., Huang, C., Arfaux, A., & Lavvas, P. 2025, ApJ — “A Multi-Species Atmospheric Escape Model with Excited Hydrogen and Helium: Application to HD 209458b.” doi:10.3847/1538-4357/ade3c9
- **Taylor, A.R.**, Dunn, A., Peacock, S., Youngblood, A., & Redfield, S. 2024, ApJ, 964, 80 — “Correlating Intrinsic Stellar Parameters with Mg II Self-reversal Depths.” doi:10.3847/1538-4357/ad22da

MANUSCRIPTS IN PREPARATION

- **Taylor, A.R.**, Koskinen, T., Huang, C., & Lavvas, P. — “Atmospheric Escape from Water-rich Sub-Neptunes.” In prep.
- Peacock, S., Husbey, L., Barker, M., **Taylor, A.R.**, Dunn, A., Barman, T.S., Hintz, D., & Shkolnik, E.L. — “PEGASUS: PHOENIX EUV Grid And Stellar Ultraviolet Spectra.” In prep, to be submitted to ApJ.

INVITED TALKS

Evaporating Worlds Workshop, Sexten, Italy *June 2026*
“Exploring atmospheric escape from water-rich sub-Neptunes with full-atmosphere models”

Stars & Planet Formation Group Meeting, University of Michigan *February 2025*

"Exploring atmospheric escape from hot-Jupiters with full-atmosphere models"

PRESENTATIONS

Exoplanets 6 Conference — Contributed Talk *June–July 2026*
"Exploring atmospheric escape from water-rich sub-Neptunes with full-atmosphere models"

American Astronomical Society — Contributed Talk *January 2026*
"Exploring atmospheric escape from hot-Jupiters and water-rich sub-Neptunes with full-atmosphere models"

Atmospheric Escape in Planetary Systems Workshop — Contributed Talk *November 2025*
"Exploring atmospheric escape from hot-Jupiters and water-rich sub-Neptunes with full-atmosphere models" Link to Talk

Solar System in Context Conference — Poster *October 2025*
"A Multi-Species Atmospheric Escape Model with Excited Hydrogen and Helium"

EPSC-DPS Joint Meeting — Contributed Talk *September 2025*
"Exploring atmospheric escape from hot-Jupiters and water-rich sub-Neptunes with full-atmosphere models" Link to Talk

Exoclimes VII Conference — Poster *July 2025*
"A Multi-Species Atmospheric Escape Model with Excited Hydrogen and Helium"

Exoplanets 5 Conference — Poster *June 2024*
"Assessment of HeI triplet absorption at 10830Å in escaping atmospheres of hot gaseous exoplanets"
Link to Poster

American Astronomical Society — iPoster *January 2023*
"Stellar chromospheres and intrinsic stellar parameters" Link to iPoster

NCSU Abstract YouTube Presentation *January 2023*
"X-ray binary Vela X-1" Link to Video

Senior Design Presentation, NCSU Physics Department *December 2022*
"Precise positioning system for a lead gamma-ray collimator" Link to Presentation

NASA Goddard Code 660 Summer Intern Symposium — Poster *August 2022*
"Stellar activity, structure, and the chromosphere"

NCSU Physics Department McCormick Undergraduate Research Symposium — Poster *May 2022*
"The Effect of Wind Speed and Roche Lobe Geometries on the Wind Dynamics of Vela X-1"

PROFESSIONAL SERVICE

NASA TESS Guest Investigator Cycle 9 Review Panel — Executive Secretary *June 9–12, 2026*
— Served as Executive Secretary for a NASA review panel.

NASA Roman 2024 Virtual Review Panel — Executive Secretary *April 21–25, 2025*
— Served as Executive Secretary for the NASA Roman 2024 virtual grant review panel.

Peer Reviewer — The Astrophysical Journal *2025–Present*
— Completed peer review of two manuscripts.

OUTREACH & SERVICE

Letters to a Pre-Scientist — Pen Pal *Aug 2025 – Present*

– Serving as a pen pal for middle schoolers to humanize STEM professionals and demystify career pathways.

Other Worlds — Curriculum Developer *Aug 2025 – Present*

– Developing an astrobiology curriculum to deliver to incarcerated youth in Tucson, AZ.

Arizona Sky School — Instructor *Feb 2025 – Present*

– Teaching K–12 students about planetary science through hands-on, outdoor learning experiences at Mt. Lemmon.
– Developed a full astrobiology day curriculum, including a JWST planets slide deck and student one-pagers.

STAR Labs — Research Mentor *Aug 2024 – Present*

– Advising a high school student's independent research project through weekly meetings and written work review.
– Mentee Tesla Lukow won First Place, the NASA Earth System Science Award, and an ISEF Finalist Trip.

Arizona Science Center — Volunteer & Girls Who STEM Mentor *Aug 2023 – Present*

– Mentoring young girls at Girls Who STEM events and participating in STEAM Nights at local elementary schools.

Miscellaneous Outreach

– Panel participant at CUWiP conferences; science talks to high school classrooms; career talks to Women in Physics clubs; field trip facilitation to Kuiper Space Sciences.